Database Management Systems: Assignment-6

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Q1. Use the Employee table from Assignment 5. Answer the followings:

EmpID	Name	Department	ManagerID	JoiningDate	Salary	City
121	Ravi Kumar	HR	201	2019-03-15	75000	Mumbai
122	Sneha Sharma	Finance	202	2021-06-10	82000	Delhi
123	Ajay Singh	IT	203	2018-09-05	95000	New York
124	Anjali Verma	HR	201	2020-02-01	68000	Pune
125	Rahul Gupta	Marketing	204	2022-01-20	55000	Toronto
126	Siddharth Jain	IT	203	2017-11-12	105000	San Francisco
127	Shivani Desai	Finance	202	2020-08-15	72000	London
128	Sunil Kumar	HR	201	2023-04-12	60000	Ahmedabad
129	Kavita Mehta	Operations	205	2019-12-18	50000	Sydney
130	Pooja Patel	IT	203	2021-09-25	88000	Chennai

a) Display Manager Id of employees whose name starts with 'A'.

Query:

SELECT ManagerID FROM employee WHERE Name LIKE 'A%';

Output:

ManagerID
203
201

b) Display book title and author, language wise.

book_id	title	author	publisher	price	language
1	AI Revolution	Andrew Ng	TechPress	750.00	English
2	Deep Learning Basics	Ian Goodfellow	MLBooks	650.00	English
3	Data Science for All	Cathy O'Neil	DataPub	400.00	English
4	Python for Beginners	Mark Lutz	CodeWorld	300.00	English
5	Machine Learning Guide	Andrew Ng	TechPress	500.00	English
6	Advanced SQL	John Smith	DBBooks	550.00	English
7	AI in Healthcare	Andrew Ng	HealthTech	800.00	English
8	C Programming	Dennis Ritchie	CodeWorld	200.00	English
9	Clean Code	Robert Martin	Prentice Hall	900.00	English
10	Java Complete Reference	Herbert Schildt	OraclePress	600.00	English
11	Blockchain Basics	Satoshi Nakamoto	CryptoBooks	1000.00	English
12	Operating Systems	Galvin	EduWorld	450.00	English
13	Database Management	Ramez Elmasri	Pearson	700.00	English
14	AI Future	Andrew Ng	TechPress	950.00	English
15	Data Mining Concepts	Han & Kamber	Pearson	550.00	English

Query:

SELECT language, title, author FROM book ORDER BY language;

Output:

language	title	author
English	AI Revolution	Andrew Ng
English	Deep Learning Basics	Ian Goodfellow
English	Data Science for All	Cathy O'Neil
English	Python for Beginners	Mark Lutz
English	Machine Learning Guide	Andrew Ng
English	Advanced SQL	John Smith
English	AI in Healthcare	Andrew Ng
English	C Programming	Dennis Ritchie
English	Clean Code	Robert Martin
English	Java Complete Reference	Herbert Schildt
English	Blockchain Basics	Satoshi Nakamoto
English	Operating Systems	Galvin
English	Database Management	Ramez Elmasri
English	AI Future	Andrew Ng
English	Data Mining Concepts	Han & Kamber

c) Display employee count department wise.

Query:

SELECT Department, COUNT(*) AS employee_count FROM employee GROUP BY Department;

Output:

department	employee_count
Finance	2
HR	3
IT	3
Marketing	1
Operations	1

d) Display all columns of book library table whose price is more than 500.

Query:

SELECT * FROM book WHERE price>500;

Output:

book_id	title	author	publisher	price	language
1	AI Revolution	Andrew Ng	TechPress	750.00	English
2	Deep Learning Basics	Ian Goodfellow	MLBooks	650.00	English
6	Advanced SQL	John Smith	DBBooks	550.00	English
7	AI in Healthcare	Andrew Ng	HealthTech	800.00	English
9	Clean Code	Robert Martin	Prentice Hall	900.00	English
10	Java Complete Reference	Herbert Schildt	OraclePress	600.00	English
11	Blockchain Basics	Satoshi Nakamoto	CryptoBooks	1000.00	English
13	Database Management	Ramez Elmasri	Pearson	700.00	English
14	AI Future	Andrew Ng	TechPress	950.00	English
15	Data Mining Concepts	Han & Kamber	Pearson	550.00	English

e) Display authors with more than 3 books publication.

Query:

SELECT author, COUNT(*) AS book_count FROM book GROUP BY author HAVING COUNT(*)>3;

Output:

author	book_count
Andrew Ng	4

f) Display publisher name and book title whose price is greater than 250, author wise.

Query:

SELECT author, publisher, title, price FROM book WHERE price>250 GROUP BY author;

Output:

author	publisher	title	price
Andrew Ng	TechPress	AI Revolution	750.00
Cathy O'Neil	DataPub	Data Science for All	400.00
Galvin	EduWorld	Operating Systems	450.00
Han & Kamber	Pearson	Data Mining Concepts	550.00
Herbert Schildt	OraclePress	Java Complete Reference	600.00
Ian Goodfellow	MLBooks	Deep Learning Basics	650.00
John Smith	DBBooks	Advanced SQL	550.00
Mark Lutz	CodeWorld	Python for Beginners	300.00
Ramez Elmasri	Pearson	Database Management	700.00
Robert Martin	Prentice Hall	Clean Code	900.00
Satoshi Nakamoto	CryptoBooks	Blockchain Basics	1000.00

g) Display author and give average price of his/her all books price.

Query:

SELECT author, AVG(price) AS avg_price FROM book GROUP BY author;

Output:

author	avg_price
Andrew Ng	750.000000
Cathy O'Neil	400.000000
Dennis Ritchie	200.000000
Galvin	450.000000
Han & Kamber	550.000000
Herbert Schildt	600.000000
Ian Goodfellow	650.000000
John Smith	550.000000
Mark Lutz	300.000000
Ramez Elmasri	700.000000
Robert Martin	900.000000
Satoshi Nakamoto	1000.000000

h) Display author and book title with highest price.

Query:

SELECT author, title, price FROM book WHERE price = (SELECT MAX(price) FROM book);

Output:

author	title	price
Satoshi Nakamoto	Blockchain Basics	1000.00

i) Display author and book title with least price.

Query:

SELECT author, title, price FROM book WHERE price = (SELECT MIN(price) FROM book);

Output:

author	title	price
Dennis Ritchie	C Programming	200.00

j) Display author and book title with second highest price.

Query:

SELECT author, title, price FROM book WHERE price = (SELECT MAX(price) FROM book WHERE price < (SELECT MAX(price) FROM book));

Output:

author	title	price
Andrew Ng	AI Future	950.00

Q2. Use the Student table from Assignment 5. Answer the followings:

RollNo	Name	Semester	DOB	AdmissionDate	HostelRoom	Branch	City
101	Suraj Shah	4	2001-05-14	2019-07-01	201	Artificial Intelligence	Mumbai
102	Sakshi Mehta	3	2002-09-20	2020-07-01	305	Information Tech	Delhi
103	Ravi Patel	4	2000-11-10	2018-07-01	202	Electronics	Ahmedabad
104	Sneha Sharma	2	2003-01-25	2021-07-01	105	Mechanical	Pune
105	Ajay Singh	4	2001-08-05	2019-07-01	402	Civil	Chennai
106	Sunil Kumar	1	2004-06-18	2022-07-01	210	Electrical	Jaipur
107	Anjali Verma	3	2002-12-30	2020-07-01	115	Chemical	Surat
108	Siddharth Jain	4	2001-03-12	2019-07-01	203	Computer Science	Lucknow
109	Rahul Gupta	2	2003-09-05	2021-07-01	312	Civil	Hyderabad
110	Shivani Desai	4	2000-10-28	2018-07-01	205	Mechanical	Indore

1. Display semester of students whose name has the letter ‘a’.

   Query:

   SELECT Semester FROM student WHERE Name LIKE '%a%';

   Output:

Semester
4
3
4
2
4
1
3
4
2
4

2. Display count of students semester wise.

   Query:

   SELECT Semester, COUNT(*) AS StudentCount FROM student GROUP BY Semester;

   Output:

Semester	StudentCount
1	1
2	2
3	2
4	5

3. Display students' names from every department whose roll number is 1.
Query:
SELECT Name, Branch FROM student WHERE RollNo=101;
Output:

```
+------------+-------------------------+
| Name       | Branch                  |
+------------+-------------------------+
| Suraj Shah | Artificial Intelligence |
+------------+-------------------------+
```

4. Display student name and semester of students who are not staying in the hostel.
Query:
SELECT Name, Semester FROM student WHERE HostelRoom IS NULL;
Output:

```
Empty set (0.001 sec)
```

5. Display student count in each semester whose birth month is August.
Query:
SELECT Semester, COUNT(*) AS StudentCount FROM student WHERE MONTH(DOB)=8 GROUP BY Semester;
Output:

```
+----------+--------------+
| Semester | StudentCount |
+----------+--------------+
|        4 |            1 |
+----------+--------------+
```

6. Display roll number and name of the student who was the first one to get admission in the college.
Query:
SELECT RollNo, Name FROM student WHERE AdmissionDate = (SELECT MIN(AdmissionDate) FROM student);
Output:

```
+--------+---------------+
| RollNo | Name          |
+--------+---------------+
|    103 | Ravi Patel    |
|    110 | Shivani Desai |
+--------+---------------+
```

7. Display the average count of students. (In any semester)
Query:
SELECT avg(StudentCount) AS AvgStudents FROM (SELECT COUNT(*) AS StudentCount FROM student GROUP BY Semester) AS x;
Output:

```
+-------------+
| AvgStudents |
+-------------+
|      2.5000 |
+-------------+
```

8. For every month (Jan-Dec) display the count of students who are having birthdays in that month.
Query:
SELECT MONTH(DOB) AS BirthMonth, COUNT(*) AS StudentCount FROM student GROUP BY MONTH(DOB) ORDER BY BirthMonth;
Output:

```
+------------+--------------+
| BirthMonth | StudentCount |
+------------+--------------+
|          1 |            1 |
|          3 |            1 |
|          5 |            1 |
|          6 |            1 |
|          8 |            1 |
|          9 |            2 |
|         10 |            1 |
|         11 |            1 |
|         12 |            1 |
+------------+--------------+
```

9. Display count of students who have taken admission in the last six months.
Query:
SELECT COUNT(*) AS RecentAdmissions FROM student WHERE MONTH(AdmissionDate) BETWEEN 7 AND 12;
Output:

```
+------------------+
| RecentAdmissions |
+------------------+
|               10 |
+------------------+
```

10. Display semester with least number of students.

Query:

SELECT Semester FROM student GROUP BY Semester ORDER BY COUNT(*)=1;

Output:

```
+----------+
| Semester |
+----------+
|        3 |
|        2 |
|        4 |
|        1 |
+----------+
```

Q3. Create a table named "Hospital management". Add the columns as per requirements.

Add the records. Answer the followings:

doctor_id	doctor_name	specialization	salary	patient_id	patient_name	admission_date	discharge_date
101	Dr. Sharma	Cardiology	60000.00	201	Anil	2025-01-10	2025-01-20
102	Dr. Mehta	Neurology	50000.00	202	Rekha	2025-02-05	2025-02-18
103	Dr. Gupta	Cancer	70000.00	203	Sanjay	2025-03-15	2025-03-30
104	Dr. Ramesh	Orthopedics	50000.00	204	Pooja	2025-04-02	2025-04-20
105	Dr. Suman	Cancer	50000.00	205	Ravi	2025-05-01	2025-05-10
106	Dr. Arora	Cardiology	50000.00	206	Mohan	2025-06-01	2025-06-15
107	Dr. Varma	Dermatology	55000.00	207	Kajal	2025-02-12	2025-02-25
108	Dr. Nair	Pediatrics	48000.00	208	Aarti	2025-03-20	2025-03-29
109	Dr. Bose	Cardiology	50000.00	209	Manoj	2025-04-10	2025-04-25
110	Dr. Iyer	Neurology	52000.00	210	Deepa	2025-05-15	2025-05-28

1. Display count of doctors as per their specialization.
   Query:
   SELECT specialization, COUNT(*) AS doctor_count FROM Hospital_Management GROUP BY specialization;
   Output:

specialization	doctor_count
Cancer	2
Cardiology	3
Dermatology	1
Neurology	2
Orthopedics	1
Pediatrics	1

2. Display doctor name from every specialization whose doctor ID is 101.
   Query:
   SELECT doctor_name, specialization FROM Hospital_Management WHERE doctor_id=101;
   Output:

doctor_name	specialization
Dr. Sharma	Cardiology

3. Display doctor ID and name who are not specialized in cancer.
   Query:
   SELECT doctor_id, doctor_name FROM Hospital_Management WHERE specialization NOT IN ('Cancer');
   Output:

doctor_id	doctor_name
101	Dr. Sharma
102	Dr. Mehta
104	Dr. Ramesh
106	Dr. Arora
107	Dr. Varma
108	Dr. Nair
109	Dr. Bose
110	Dr. Iyer

4. Display doctors count in each specialization whose salary is 50000.
   Query:
   SELECT specialization, COUNT(*) AS doctor_count FROM hospital_management WHERE salary=50000 GROUP BY specialization;
   Output:

specialization	doctor_count
Cancer	1
Cardiology	2
Neurology	1
Orthopedics	1

5. Display patient ID and patient name who was the first one got admitted in the hospital.
   Query:
   SELECT patient_id, patient_name FROM Hospital_Management WHERE admission_date = (SELECT MIN(admission_date) FROM Hospital_Management);
   Output:

patient_id	patient_name
201	Anil

6. Display the average count of patients. (In any month)
   Query:
   SELECT AVG(*) AS avg_patients FROM (SELECT MONTH(admission_date) AS month_no, COUNT(*) AS patient_count FROM Hospital_Management GROUP BY MONTH(admission_date)) AS monthly;
   Output:

avg_patients
1.6667

7. For every month (Jan-may) display the count of patients who got discharged in that month.
   Query:
   SELECT MONTH(discharge_date) AS month_no, COUNT(*) AS discharged_patients FROM Hospital_Management WHERE MONTH(discharge_date) BETWEEN 1 AND 5 GROUP BY MONTH(discharge_date);
   Output:

month_no	discharged_patients
1	1
2	2
3	2
4	2
5	2

8. Display count of patients who got admitted in the last six months.
   Query:
   SELECT COUNT(*) AS recent_patients FROM Hospital_Management WHERE MONTH(admission_date) BETWEEN 7 AND 12;
   Output:

recent_patients
0

9. Display specialization with least number of doctors.

Query:

SELECT specialization FROM Hospital_Management GROUP BY specialization ORDER BY COUNT(*)=1;

Output:

```
+----------------+
| specialization |
+----------------+
| Neurology      |
| Cardiology     |
| Cancer         |
| Dermatology    |
| Orthopedics    |
| Pediatrics     |
+----------------+
```